

Deliverable 11 – Executive Recommendation Report

Healthcare Patient Experience Analytics using Power BI

Introduction

Based on the analysis of HCAHPS survey data and the interactive Power BI dashboards, several opportunities for improving patient experience and healthcare quality were identified. This report provides strategic recommendations to help the National Healthcare Quality Commission (NHQC), healthcare administrators, and policy makers enhance patient satisfaction, strengthen healthcare service delivery, and support continuous quality improvement.

The recommendations are aligned with the analytical findings obtained from national, regional, state, and healthcare quality measure analysis.

Executive Recommendations

1. Improve Care Transition Programs

Observation

Care Transition received the lowest patient satisfaction score among all healthcare quality measures.

Recommendation

Develop standardized care transition protocols that ensure patients receive clear discharge plans, follow-up appointments, medication guidance, and post-discharge support.

Expected Benefit

- Better continuity of care
- Reduced hospital readmissions
- Higher patient satisfaction

2. Strengthen Communication About Medicines

Observation

Patient satisfaction regarding communication about medicines is lower than communication with doctors and nurses.

Recommendation

Introduce standardized medication counseling before patient discharge and provide simplified written medication instructions.

Expected Benefit

- Improved medication adherence
 - Reduced patient confusion
 - Better treatment outcomes
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3. Increase Patient Survey Participation

Observation

The survey response rate is approximately **26.61%**, indicating relatively low participation.

Recommendation

Implement digital surveys, SMS reminders, QR code-based feedback forms, and patient awareness campaigns to encourage greater survey participation.

Expected Benefit

- More representative patient feedback
 - Improved reliability of performance analysis
 - Better-informed decision-making
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4. Reduce Regional Performance Gaps

Observation

Healthcare performance varies significantly across different U.S. regions.

Recommendation

Identify best practices from high-performing regions and implement standardized quality improvement programs in lower-performing regions.

Expected Benefit

- More consistent healthcare quality
 - Reduced regional disparities
 - Improved nationwide patient satisfaction
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5. Benchmark High-Performing States

Observation

Several states consistently outperform others in patient satisfaction.

Recommendation

Document operational best practices from top-performing states and share them with lower-performing healthcare systems through knowledge-sharing initiatives.

Expected Benefit

- Faster organizational learning
 - Improved healthcare quality across all states
 - Better resource utilization
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6. Improve Hospital Environment

Observation

Hospital environment-related measures, including quietness, perform below several communication-related measures.

Recommendation

Invest in noise reduction initiatives, improve ward management practices, and strengthen environmental quality standards.

Expected Benefit

- Improved patient comfort
 - Better recovery experience
 - Higher patient satisfaction
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7. Monitor Satisfaction Trends Continuously

Observation

Patient satisfaction improved until 2021 before declining during the most recent reporting periods.

Recommendation

Establish quarterly performance reviews using Power BI dashboards to identify declining trends early and implement corrective actions.

Expected Benefit

- Faster issue identification
 - Continuous quality improvement
 - Improved long-term healthcare performance
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8. Expand KPI-Based Performance Monitoring

Observation

KPI-driven dashboards simplify healthcare performance evaluation.

Recommendation

Introduce additional KPIs related to patient safety, hospital efficiency, emergency care performance, and clinical outcomes.

Expected Benefit

- Comprehensive healthcare performance monitoring
 - Better strategic planning
 - Enhanced executive reporting
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9. Promote Data-Driven Decision Making

Observation

Interactive dashboards provide real-time healthcare insights across multiple dimensions.

Recommendation

Encourage healthcare executives and administrators to use Power BI dashboards during operational reviews and strategic planning meetings.

Expected Benefit

- Faster decision-making
 - Increased transparency
 - Improved organizational performance
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10. Enhance Future Analytics Capabilities

Observation

The current solution focuses primarily on descriptive and diagnostic analytics.

Recommendation

Expand future analytics capabilities by integrating predictive analytics, machine learning models, hospital operational data, and real-time patient monitoring systems.

Expected Benefit

- Predictive healthcare insights
 - Early identification of healthcare risks
 - Improved patient outcomes
 - More proactive healthcare management
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Strategic Roadmap

Phase	Recommendation	Priority
Phase 1	Improve Care Transition	High
Phase 1	Increase Survey Participation	High
Phase 1	Improve Communication About Medicines	High
Phase 2	Reduce Regional Performance Gaps	Medium
Phase 2	Benchmark High-Performing States	Medium
Phase 2	Improve Hospital Environment	Medium
Phase 3	Expand KPI Monitoring	Medium
Phase 3	Promote Data-Driven Decision Making	Medium
Phase 4	Implement Predictive Analytics	Long-Term

Expected Business Outcomes

Implementation of these recommendations is expected to:

- Improve patient satisfaction across hospitals.
 - Strengthen communication between patients and healthcare providers.
 - Increase survey participation and data reliability.
 - Reduce regional variations in healthcare quality.
 - Improve benchmarking across states.
 - Support evidence-based healthcare policy development.
 - Enhance operational efficiency through Business Intelligence.
 - Enable continuous monitoring of healthcare performance.
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Conclusion

The Healthcare Patient Experience Analytics project demonstrates how Business Intelligence can transform healthcare survey data into actionable insights. The recommendations presented in this report focus on improving patient experience, strengthening healthcare quality, reducing performance gaps, and promoting data-driven decision-making. By implementing these strategic initiatives, healthcare organizations can enhance service quality, improve patient outcomes, and establish a culture of continuous performance improvement supported by interactive analytics and KPI-driven reporting.

